

PROJECT REPORT

Project Report: Particle in a Box Simulation

1. Introduction

The particle in a box model is one of the most fundamental problems in quantum mechanics. It describes a particle that is confined within an infinitely deep potential well, where it can only occupy discrete energy levels.

This project implements a numerical simulation of this system using Python. The goal is to compute and visualize the wavefunctions and corresponding energy levels for different quantum states.

2. Objectives

- To simulate the one-dimensional quantum system of a confined particle

- To compute normalized wavefunctions for different quantum numbers
 - To calculate quantized energy levels
 - To visualize the results using graphical plots
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3. Theoretical Background

The system is based on the particle in a box solution to the Schrödinger equation.

3.1 Wavefunction

The normalized wavefunction is given by:

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$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

\$\$

Where:

- (n) = quantum number (1, 2, 3, ...)
- (L) = length of the box

3.2 Energy Levels

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$$E_n = \frac{n^2 h^2}{8mL^2}$$

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Where:

- (h) = Planck's constant
- (m) = mass of the particle

4. Methodology

4.1 Tools and Libraries

The simulation is implemented using:

- Python
- NumPy (for numerical computations)
- Matplotlib (for plotting)

4.2 Implementation Steps

1. Define Constants
 - Planck's constant (h)
 - Electron mass (m)
 - Box length (L)

2. Discretize the Space

- Create an array of position values from 0 to (L)

3. Define Functions

- Wavefunction function ($\psi_n(x)$)
- Energy function (E_n)

4. Plot Results

- Loop through quantum numbers ($n = 1$ to 4)
- Plot wavefunctions on the same graph
- Label each curve with its energy value

5. Results

The simulation produces a graph showing wavefunctions for different quantum states:

- ($n = 1$): Single peak, no internal nodes
- ($n = 2$): One node
- ($n = 3$): Two nodes

- ($n = 4$): Three nodes

Observations:

- The number of nodes increases with quantum number
 - Wavefunctions are zero at the boundaries
 - Energy increases rapidly with increasing (n)
 - Higher energy states show more oscillatory behavior
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6. Discussion

The results align with theoretical predictions:

- Energy levels follow a quadratic relationship with (n)
- The confinement leads to quantization, unlike classical systems
- The shape of wavefunctions reflects standing wave patterns

This simulation demonstrates how quantum systems differ fundamentally from classical mechanics, especially in terms of energy discretization.

7. Applications

The particle in a box model is important in:

- Nanotechnology
 - Quantum dots
 - Semiconductor physics
 - Molecular systems
 - Understanding electron confinement
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8. Limitations

- Assumes infinite potential walls (idealized system)
- No time-dependent behavior included

- Does not account for external forces or interactions
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9. Future Improvements

- Add time-dependent Schrödinger equation visualization
 - Plot probability density ($|\psi(x)|^2$)
 - Convert energy values into electron volts (eV)
 - Create an interactive interface (e.g., sliders for (n) , (L))
 - Extend to 2D or 3D systems
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10. Conclusion

This project successfully simulates the particle in a box model and visualizes its key properties. It provides a clear demonstration of quantized energy levels and wavefunction

behavior, reinforcing core concepts in quantum mechanics.

11. References

- Standard Quantum Mechanics textbooks
 - Schrödinger equation solutions
 - Educational physics resources
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